DESIGN AND DEVELOPMENT

Revisions

19	19/10/2023	**HSEQ**	**QA-HSEM** C.Delli Qquadri M.Tremolada	**MD** P.Vacis	Logo modified, Trillium Pumps Italy references removed and use extended to the other plants of the organization
18	05/05/2023	R&D Mgr M.Borghetti	QA/HSEM C.Delli Quadri/ M.Tremolada	MD E.Garibotti	General update
17	25/02/2021	R&D-ENG D R.Ciceri	QA/HSEM C.Delli Quadri/ M.Tremolada	MD P.Macchi	Addition of paragraph 6
16	30/10/2020	R&D-ENG D R.Ciceri	QAM/HSEM S.Gobbi / M.Tremolada	MD P.Macchi	Transition to ISO 45001
15	5/04/2018	R&D-ENG D R.Ciceri	QAM F.Rizzi	MD P.Macchi	Addition of Mod.330 and use of different forms clarified. Transition to ISO 9001:2015. Addition of ATEX, PED, Machinery Directive and Product Stewardship references
Rev.	*Date*	*Drawn up*	*Checked*	*Approved*	*Reason*

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Contents

1 Scope and field of application 3

2 Acronyms 3

3 Bodies involved 3

4 The NPI process 4

4.1 Phase 0: Discovery 5

4.2 Phase 1: Scoping 5

4.3 Phase 2: Feasibility 5

4.4 Phase 3: Development 6

4.5 Phase 4: Qualification 6

4.6 Phase 5: Launch 6

5 Operating modes 6

5.1 Design of new families of pumps 6

5.2 Design to order 7

5.2.1 Execution of alternative constructions for existing hydraulics 7

5.2.2 Completion of existing families with the addition of new hydraulics 8

5.2.3 Completion of existing families with the addition of hydraulics scaled up or down from validated geometries 9

5.2.4 Execution of mechanical configurations outside the operating range of the “pre-configured” product 9

6 Patents 11

6.1 Registration of new patents 11

6.2 Renewal of filing of a patent 11

7 Flow charts 11

8 References 16

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1 Scope and field of application

The scope of this procedure is to describe the design activities of new products intended as:

- development of new families of pumps;
- completion of existing families with the addition of new hydraulics;
- completion of existing families with the addition of hydraulics scaled up or down from validated geometries;
- execution of alternative constructions for existing hydraulics;
- execution of mechanical configurations for existing hydraulics but that are outside the operating range of the “pre-configured” product.

Products are defined as “new” when they require relevant design, estimate and test activities.

2 Acronyms

CAPEX: Capital Expenditure

CFD: Computational Fluid Dynamics

ENG: Engineering

FEM: Finite Element Method

HT: Hydrotest

KOM: Kickoff Meeting

NPI: New Product Introduction

NPP: New Product Proposal

PDS: Pump Data Sheet

PT: Performance Test

R&D: Research and Development

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3 Bodies involved

The new product introduction (NPI) process involves various company functions in various capacities:

- **SALES** is responsible for bringing the "voice of the customer" to the company, i.e., for identifying market opportunities for which the development of new products or the evolution of existing products is required. The activity firstly consists of analyzing Customers requests and defining wheTther or not there is already a standard product (performance curve already existing) for the application, but in fact choices must be made following market analysis, supported by objective data.
- **PRODUCT MANAGEMENT** is responsible for all activities related to managing the product during the whole of its life cycle, in compliance with the predetermined objectives and in coordination with the other company functions.
- **ENG** is responsible for defining the project specifications and for verifying that the new product complies with these and with all EC regulations in force, such as: Machinery Directive, ATEX and PED where applicable, as well as with the guidelines defined by organization. As project contact, it must also guarantee development times and involve the other company functions as required.
- **R&D** is responsible for the development of new families of pumps, for designing new hydraulics and for advanced mechanical calculations. In parallel, it defines the guidelines for the development and implementation of new technologies.
- **QUALITY ASSURANCE** must support ENG/R&D in the design in accordance with the international regulations of reference.
- **MANUFACTURING** must support the validation phases, guaranteeing the spaces necessary for the construction of any prototypes required, for carrying out tests and must ensure feasibility, contributing to the assessment of the estimated costs.
- **SUPPLY CHAIN** must contribute to the definition of the make or buy logics aimed at ensuring costs in line with market demands.
- **LEGAL** must perform the checks required for the protection of intellectual property.

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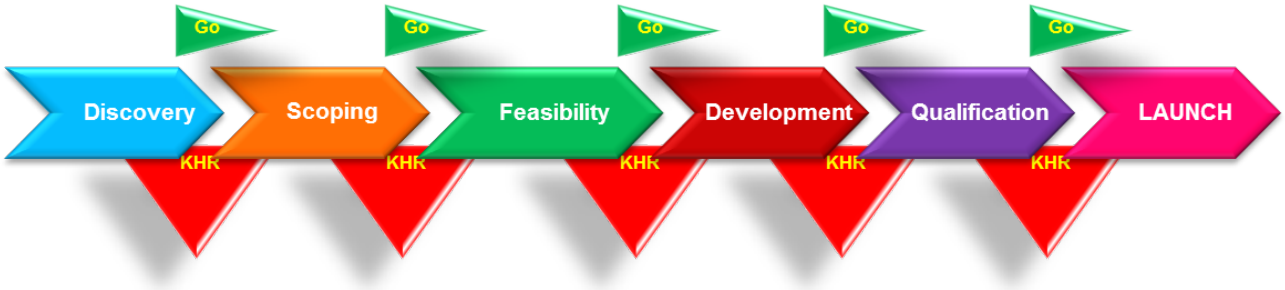
4 The NPI process

New products must be introduced onto the market in accordance with the instructions of the New Product Introduction process based on 6 development phases, spaced by decision points (phase gates), during which the project is collectively assessed in order to define whether to continue according to the strategy:

- **GO**: the project is proceeding in line with the specifications and is ready to move to the next phase.
- **KILL**: development is shut down as the project does not satisfy the requirements or the market conditions that were referred to have changed.
- **HOLD**: the project remains valid but is less interesting than others in terms of return on investment, and is therefore put on hold, waiting for the necessary resources to become available.
- **RECYCLE**: the project remains interesting but does not meet all the specifications and therefore a return to the previous phase is necessary.



4.1 Phase 0: Discovery

Identification of new ideas and market opportunities

Reference documents:

- NPP - New Product Proposal (Mod.311)
- NPI - Contract (Mod.312)
- Gate Review Checklist - Gate 1 (Mod.314)

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4.2 Phase 1: Scoping

Definition of the project perimeter, verifying market opportunities, risks and potential benefits linked to development of the new product identified. Analysis must be fast and at zero cost, in this phase no investments are yet planned.

Reference documents:

- Gate Review Checklist - Gate 2 (Mod.315)

4.3 Phase 2: Feasibility

Verification of the technical feasibility and detailed project definition, which must include at the very least:

- Technical definition of the product
- Economic analysis with definition of any CAPEX
- Definition of the project plan

Reference documents:

- CAPEX (Mod.110)
- NPI Technical Contract (Mod.313)
- Gate Review Checklist - Gate 3 (Mod.316)

4.4 Phase 3: Development

Complete project development as regards technical development, analysis of costs, production problems and quality.

Reference documents:

- Gate Review Checklist - Gate 4 (Mod.317)
- Constructed as per NPI technical contract (Mod.313)

4.5 Phase 4: Qualification

Technical validation of the project obtained through passing all the necessary tests and definition of the production plans and supply processes.

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Reference documents:

- Gate Review Checklist - Gate 5 (Mod.318)
- Pump Qualification (Mod.330)

4.6 Phase 5: Launch

Launching the product on the market.

Reference documents:

- Gate Review Checklist - Gate 6 (Mod.319)

5 Operating modes

5.1 Design of new families of pumps

The term “New Product Design” refers to the development activities of new products designed by organization even if not directly requested in offers or open orders. In these cases, the complete NPI process is followed, starting from identification of the opportunity, the result of market analysis and/or of the “no bids” tracked by SALES. (**Review of Design inputs and Design controls**: see Chap. 4).

In addition to following the complete NPI process, this case differs from the those above by the presence of a prototype, useful for the project validation phase, and by a more structured launch phase, required to introduce the new product to the sales force and to promote it on the market (**Validation of the design outputs**).

The project requires the drafting of a “new product proposal” which must be undersigned by all company functions through the form of an “NPI contract”. A project of this type typically requires a dedicated CAPEX, to be prepared in due time.

Notes: *The operating mode is summarized by the Flow Chart “B”, in paragraph 6 “Flow Charts”.*

The design of a new family of pumps is the only case of “complete” NPI and therefore each phase must be formally closed through the use of the relevant checklists.

Taking account of the indications received from SALES and PRODUCT MANAGEMENT, the design process of a new family of pumps is the responsibility of R&D supported by ENG in the Qualification and Product launch phases.

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5.2 Design to order

The term "Design to order" refers to the development activities of a "new hydraulics" or of addition to hydraulics scaled up or down from previously validated geometries for completion of a range of products or to the design of a new version of existing hydraulics (typically design of bodies different from the standard Top-Top), or to the execution of mechanical configurations for existing hydraulics but outside the operating range of the "pre-configured" product.

In this case, ENG or R&D are in charge of the design to order process, either completely or partially, depending on the responsibility assignment matrix and on the specific indications in the various phases.

5.2.1 Execution of alternative constructions for existing hydraulics

Review of design inputs [ENG]: The request for a body with outlets in a non-existing execution or for other "pre-configured" special executions typically takes place downstream of the order acquisition based on the information present on the PDS and shared in the order KOM. In this case, the NPI process starts from the development phase, as feasibility is taken for granted: all the options present in the selection programme are feasible but are only produced on an order basis.

Design controls [ENG]: In these cases, the existing drawings are modified and/or new drawings are issued downstream of the mechanical checks through finite element analysis. Fluid dynamic checks through CFD analysis are not generally necessary.

Validation of the design outputs [ENG]: project validation takes place with HT and PT carried out on the pump constructed to order. No launch phase is required.

Notes: *The operating mode is summarized by the Flow Chart "C", in paragraph 6 "Flow charts".*

For this case, completion of the checklists is not required, but only monitoring of issue of the drawings required to close the "Development" phase and tracking of the positive outcome of the performance test through the form 330 "Pump Qualification".

5.2.2 Completion of existing families with the addition of new hydraulics

Review of design inputs [ENG/ R&D]: The request for new hydraulics, or for special executions that are not "pre-configured", typically takes place during the offer phase when ENG/R&D is activated by SALES with a web ticket. In these cases, the NPI process starts from the feasibility phase, i.e., technical and economic assessment of the product is carried out, tracked through a new NPI project.

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After feasibility has been verified, formalized through the closing of gate 3, the product can be offered but development of the project remains on-hold until order acquisition, or until indication to proceed agreed with the management.

In the case of new hydraulics, R&D designs the basic hydraulics and publishes the relevant curve/technical data essential for the offer to continue, defining at least the following fundamental parameters:

- Impeller eye calculation
- Impeller Outlet calculation.
- Inlet angles.
- Estimate of dimensions (flange diameters, non-definitive overall dimensions) and weights.

Design controls [R&D]: Case by case there is the opportunity to carry out preliminary CFD analysis or to proceed with detailed design only after order acquisition. In any case, CFD and FEM analysis are carried out during the Development phase.

Validation of the design outputs [ENG]: No prototypes are required and therefore validation takes place with the tests established for the order and launch consists in making the new hydraulics available for all existing offers.

Notes: *The operating mode is summarized by the Flow Chart "A", in paragraph 6 "Flow charts".*

The release of new hydraulics is tracked through form 313 Technical Contract; it is not necessary to complete the other checklists. The result of the performance test is tracked through completion of form 330 "Pump Qualification".

5.2.3 Completion of existing families with the addition of hydraulics scaled up or down from validated geometries

Review of design inputs [ENG]: The request for new hydraulics scaled up or down from previously validated geometries typically takes place during the offer phase when ENG is activated by SALES with a web ticket. In these cases, the NPI process starts from the feasibility phase, i.e., technical and economic assessment of the product is carried out, tracked through a new NPI project. After feasibility has been verified, formalized through the closing of gate 3, the product can be offered but development of the project remains on-hold until order acquisition, or until indication to proceed agreed with the management.

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In the case of hydraulics scaled up or down from previously validated geometries, ENG publishes the relevant curve/technical data essential for the offer to continue, defining at least some fundamental parameters (flange diameters, non-definitive overall dimensions) and weights of the components.

Design controls [ENG]: In these cases, new drawings are issued downstream of the up- or down-scaling defined in the previous phase. Generally, in this type of activity fluid dynamic checks through CFD analysis and/or mechanical checks through finite element analysis are not necessary.

Validation of the design outputs [ENG]: No prototypes are required and therefore validation takes place with the tests established for the order and launch consists in making the new hydraulics available for all existing offers.

The release of new hydraulics is tracked through form 313 Technical Contract; it is not necessary to complete the other checklists. The result of the performance test is tracked through completion of form 330 "Pump Qualification".

5.2.4 Execution of mechanical configurations outside the operating range of the "pre-configured" product

Review of design inputs [ENG/ R&D]: The request for a mechanical configuration or for other special executions outside the operating range of the "pre-configured" product typically takes place during the offer phase when ENG/R&D is activated by SALES with a web ticket. In these cases, the NPI process starts from the feasibility phase, i.e., technical and economic assessment of the product is carried out, tracked through a new NPI project. After feasibility has been verified, formalized through the closing of gate 3, the product can be offered but development of the project remains on-hold until order acquisition, or until indication to proceed agreed with the management.

Design controls [R&D]: In these cases, the existing drawings are modified and/or new drawings are issued downstream of the mechanical checks through finite element analysis. Fluid dynamic checks through CFD analysis are not generally necessary.

Validation of the design outputs [ENG]: No prototypes are required and therefore validation takes place with the tests established for the order and launch consists in making the new hydraulics available for all existing offers, thereby expanding the operating range of the "pre-configured" product.

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The release of a new mechanical configuration outside the operating range of the "pre-configured" product is tracked through form 313 Technical Contract; it is not necessary to complete the other checklists. The result of the performance test is tracked through completion of form 330 "Pump Qualification".

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6 Patents

6.1 Registration of new patents

ENG and R&D functions must notify R&D Manager of all potential new inventions/innovations; the latter will assess whether to register the product/technology, based on its features and following preliminary analysis of competitors on the actual existence of characteristics of novelty.

Final approval of the decision for registration is the responsibility of the Managing Director who, with the support of the Legal Affairs function, will assess whether to involve an outside consultant for the clearance process and to proceed with the international registration process.

6.2 Renewal of filing of a patent

In the event of deciding to confirm and renew filing of a patent it is also necessary to carry out an internal verification on the advantages deriving from maintaining its registration; the considerations of the R&D Manager are shared with the Managing Director, who is responsible for making the final decision.

7 Flow charts

The flow charts used for design in the most typical situations described above are provided below.

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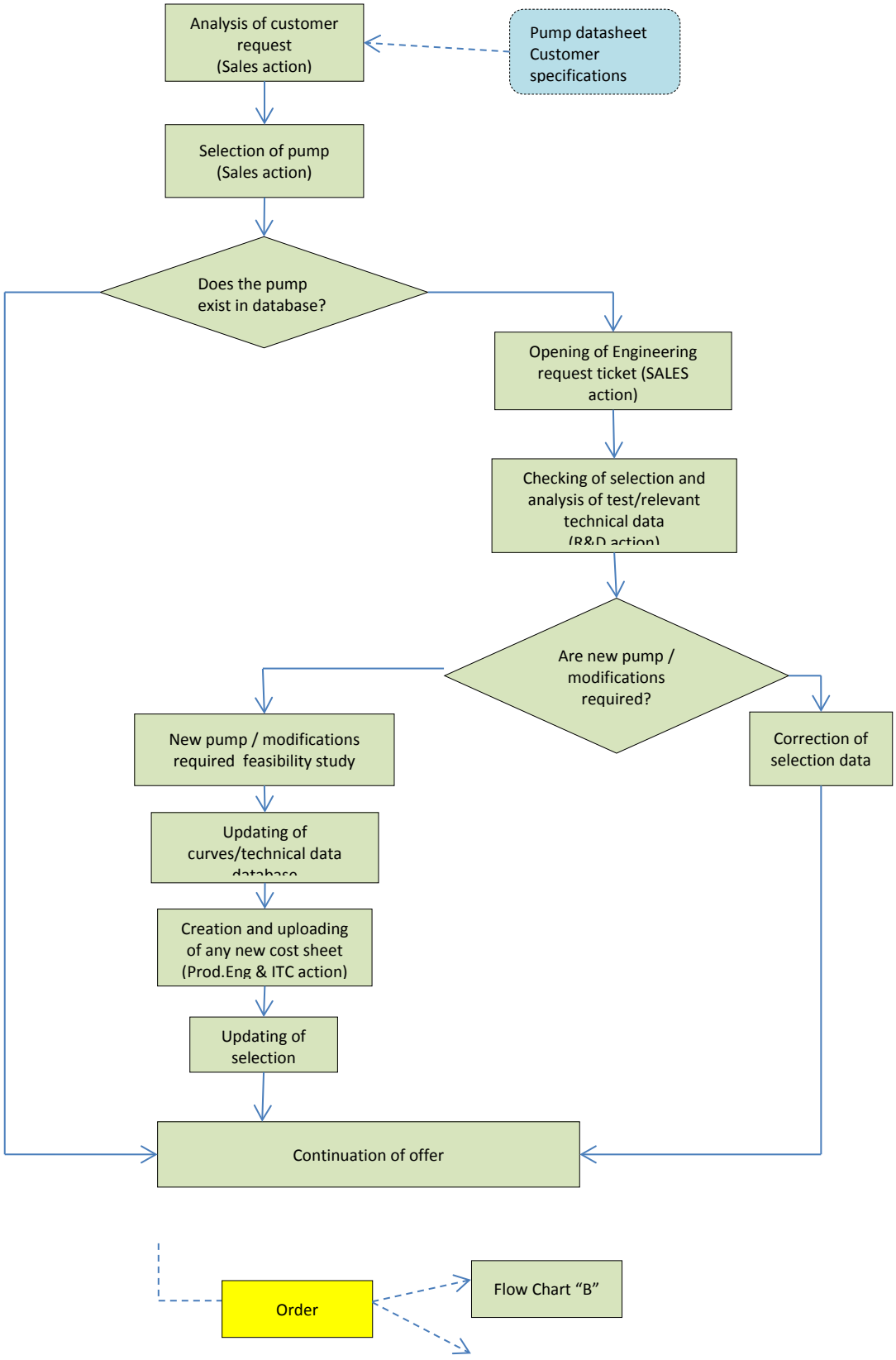
- *Flow Chart "A": Design to order – Completion of existing families with the addition of new hydraulics*



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Flow Chart "C"

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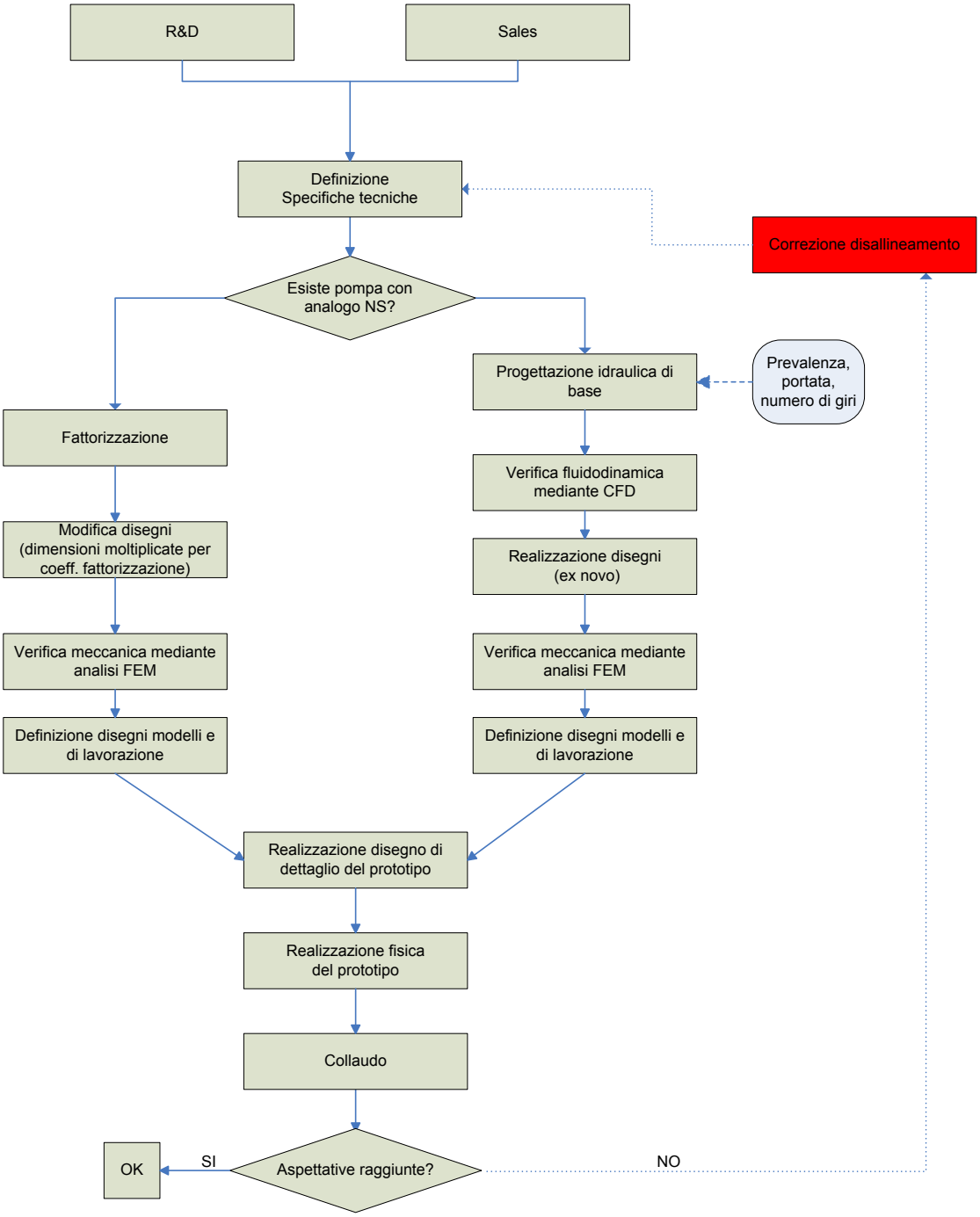
- *Flow Chart "B": Design of new families of pumps*



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Definizione specifiche tecniche - Definition of technical specifications

Correzione disallineamento – Correction of misalignment

Esiste pompa con analogo NS? - Does a pump with the same NS exist?

Fattorizzazione - Factorization

Modifica disegni (dimensioni moltiplicate per coeff. Fattorizzazione) – Modification of drawings (dimensions multiplied by factorization coeff.)

Verifica meccanica mediante FEM analisi – Mechanical checks through FEM analysis

Definizione disegni modelli e di lavorazione – Definition of model and machining drawings

Progettazione idraulica di base – Design of basic hydraulics

Prevalenza, portata, numero di giri – Head, flow rate, rpm

Verifica fluidodinamico mediante CFD – Fluid dynamic checks through CFD

Realizzazione disegni (ex novo) – Production of drawings (ex novo)

Verifica meccanica mediante FEM analisi – Mechanical checks through FEM analysis

Definizione disegni modelli e di lavorazione - Definition of model and machining drawings

Realizzazione disegno di dettaglio del prototipo – Production of detailed drawing of prototype

Realizzazione fisica del prototipo – Physical production of prototype

Collaudo - Testing

Aspettative raggiunte? – Have expectations been reached?

Si – Yes

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Flow chart "C":

Design to order – Execution of alternative constructions for existing hydraulics

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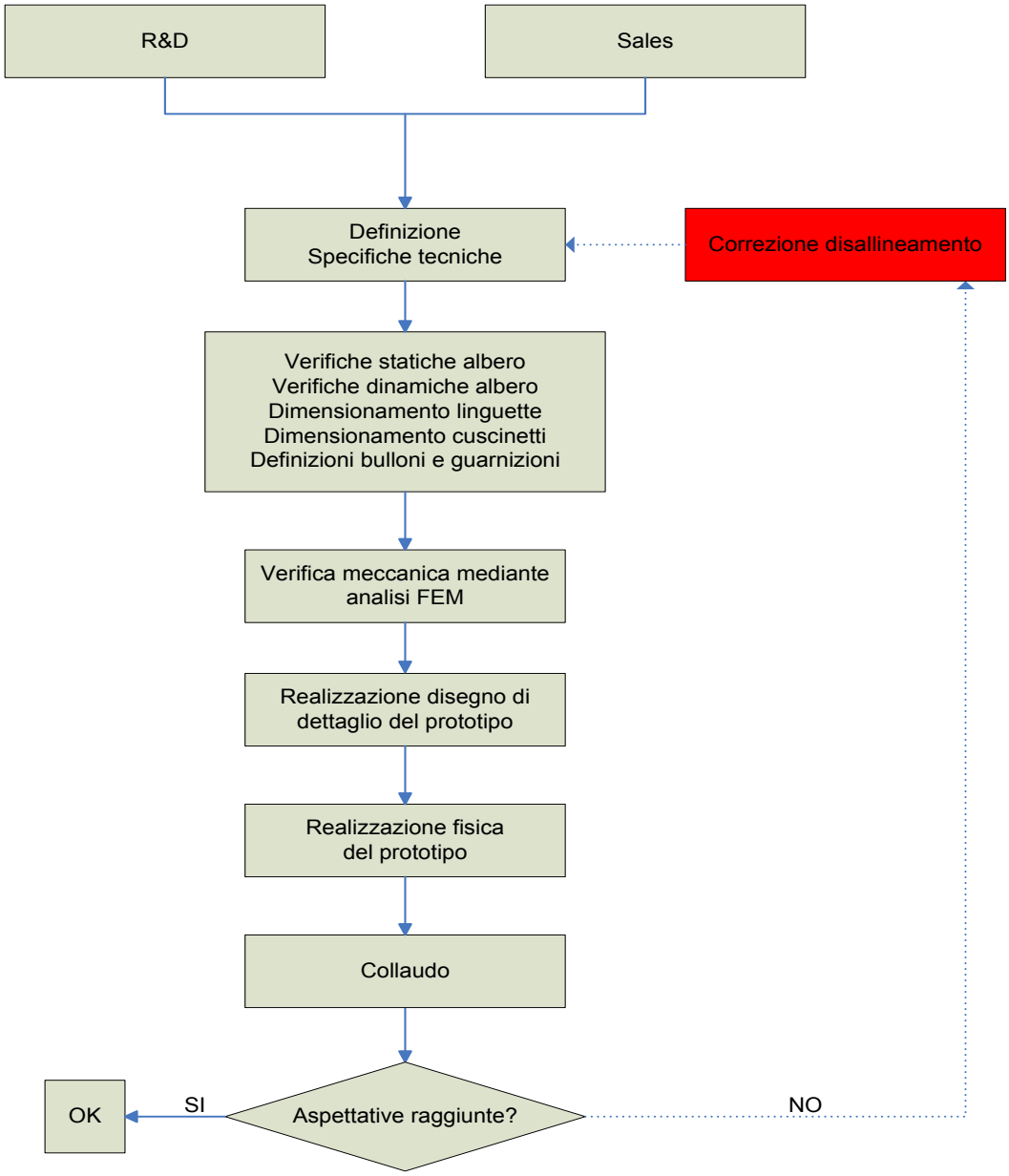

Definizione specifiche tecniche - Definition of technical specifications

Correzione disallineamento – Correction of misalignment

Verifiche statiche albero –Static tests on shaft

Verifiche dinamiche albero – Dynamic tests on shaft

Dimensionamento linguette – Key dimensioning

Dimensionamento cuscinetti – Bearing dimensioning

Definizione bulloni e guarnizioni – Definition of bolts and seals

Verifica meccanica mediante FEM analisi – Mechanical checks through FEM analysis

Realizzazione disegno di dettaglio del prototipo – Production of detailed drawing of prototype

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Realizzazione fisica del prototipo – Physical production of prototype

Collaudo - Testing

Aspettative raggiunte? – Have expectations been reached?

Si – Yes

8 References

- UNI EN ISO 9001 / UNI EN ISO 14001 / UNI ISO 45001
- API 610

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